

Pricing Game under Search Prominence Asymmetry

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1 Extended Abstract

This paper reports a laboratory experiment on the welfare effects of position auctions when consumer search is endogenous. Two ex ante identical firms bid for a prominent position that is costless for the consumer to inspect, while inspecting the second position requires paying a sunk search cost. After positions are allocated, firms simultaneously post prices that are observed by the consumer. The consumer first inspects the prominent firm and learns her match value with it. She then decides whether to incur the search cost to inspect the non-prominent firm, which reveals her (initially hidden) match value. Search features free recall. Match values are i.i.d. binary across firms (the consumer either likes a product or not), and the consumer can exit at any stage for an outside option normalized to zero.

The consumer’s search decision depends on the observed price gap and on her realized match with the prominent firm. The pricing subgame has no pure-strategy equilibrium: prices affect whether the consumer searches, generating undercutting incentives to deter or trigger search. I characterize a unique asymmetric mixed-strategy equilibrium pinned down by indifference and boundary conditions. A distinctive implication is that the non-prominent firm places an atom at its lower price bound, interpreted as a commitment to a “fire sale.” Moreover, the prominent firm’s price distribution first-order stochastically dominates the non-prominent firm’s, implying that prominence confers market power. This contrasts with prominence models in which the top-ranked firm prices more aggressively to convert initial attention into immediate sales.

The experiment tests these predictions by varying search costs, price transparency, and the position-allocation rule (second-price auction versus random assignment). I measure bids, prices, search and stopping behavior, and welfare, and assess systematic deviations from equilibrium predictions. The design also allows tests of behavioral search (e.g., quantal-response search). Such deviations can feed into pricing and bidding through backward induction. The results speak to platform design and regulation of paid prominence and price disclosure, with implications for sponsored-search mechanisms such as Google’s Generalized Second Price (GSP) auction and for antitrust policy.

2 The Model Environment

We consider a market with two firms, indexed by $j \in \{1, 2\}$, and a representative, risk-neutral consumer. The firms sell horizontally differentiated products with zero marginal cost. The ex-ante quality of the firms are the same, shutting down a potential role of prices in signaling quality differences. The interaction unfolds in two stages:

1. **Stage 1 (Position Auction):** Firms bid for prominence in a Generalized Second Price (GSP) auction. The firm with the higher bid wins Position 1 (the prominent slot) and pays the loser's bid. The loser takes Position 2 and pays 0. Here with $N = 2$ firms, GSP is equivalent to the standard Second Price auction. This stage endogenously determines the identity of Firm 1 (the winner) and Firm 2 (the loser). Firms' valuation of the top slot is also endogenously shaped by their belief of how the pricing game will be played out.
2. **Stage 2 (Pricing Game):** Having been assigned positions, firms simultaneously set prices p_1 and p_2 . Consumers observe these prices (transparent pricing) and engage in sequential search.

The remaining of the essay focuses on the derivation of the equilibrium in the Stage 2 pricing subgame.

2.1 Consumer Search and Demand

Consumers have unit demand. The match value of product j for a consumer, denoted V_j , is an independent (across firms and consumers) and identically distributed random variable drawn from a binary distribution:

- $V_j = V$ with probability q .
- $V_j = 0$ with probability $1 - q$.

for some $q \in (0, 1)$. Consumers have an outside option of 0. Firm 1 is the “default” option; consumers observe Firm 1's match value V_1 and price p_1 at a search cost of 0. After observing Firm 1, the consumer can choose to search Firm 2. Searching Firm 2 incurs a cost $s > 0$ and reveals V_2 . This information cost is sunk in the sense that even if consumers do not end up buying from Firm 2, they cannot recover this cost. Assume $s < qV$ throughout. Prices p_1 and p_2 are advertised and observed prior to any search decision. Assume free recall, i.e. consumers can always go back to buy from Firm 1 after examining Firm 2 without additional cost. Consumers' utility function if buying from Firm j is:

$$u_j = V_j - p_j - s \mathbb{1}_{\text{searched } j=2} \tag{1}$$

Search Condition. The consumer's decision to search Firm 2 depends on the realization of V_1 and the observed prices:

1. **If $V_1 = 0$:** The consumer searches Firm 2 if the expected gain exceeds the cost: $q(V - p_2) \geq s$. This imposes a reservation price on Firm 2's pricing: $p_2 \leq V - \frac{s}{q}$. (Reservation price for Firm 1 is straightforward, $p_1 \leq V$.)
2. **If $V_1 = V$:** The consumer holds a current utility of $V - p_1$. Searching Firm 2 yields a gain only if $V_2 = V$ and $p_2 < p_1$. The expected net gain is $q(p_1 - p_2) - s$. Thus, the consumer searches if and only if the following search condition holds:

$$p_1 - p_2 \geq \frac{s}{q}. \quad (2)$$

Remark 1 (Tie-Breaking at Indifference). *The consumer is indifferent when $p_1 - p_2 = \frac{s}{q}$. In the Mixed Strategy Nash Equilibrium (MSNE) derived below, firms independently mix over continuous distributions (with at most one mass point). Consequently, the probability of the price pair (p_1, p_2) landing exactly on the knife-edge indifference condition is zero. Therefore, the specific tie-breaking rule adopted by consumers does not affect the expected profit functions or the derivation of the equilibrium. Equation (2) breaks the indifference in favor of Firm 2.*

2.2 Firm Profit Functions

Based on the consumer search behavior, we define the demand and profit functions. Firm 1 acts as a monopolist for consumers who match with it, unless its price is high enough to trigger search. The free-recall assumption ensures Firm 1 will get some returning demand from its matched consumers when search is activated. Firm 2 acts as a competitor for switchers but enjoys local monopoly power over consumers who found $V_1 = 0$, called the niche demand.

The expected profits, π_1 and π_2 , conditional on prices, are defined by two regimes:

$$\pi_1(p_1, p_2) = \begin{cases} p_1 \cdot q & \text{if } p_1 < p_2 + \frac{s}{q} \quad (\text{Search Deterred}) \\ p_1 \cdot q(1 - q) & \text{if } p_1 \geq p_2 + \frac{s}{q} \quad (\text{Search Activated}) \end{cases} \quad (3)$$

$$\pi_2(p_1, p_2) = \begin{cases} p_2 \cdot q(1 - q) & \text{if } p_2 > p_1 - \frac{s}{q} \quad (\text{Search Deterred}) \\ p_2 \cdot [q(1 - q) + q^2] = p_2 \cdot q & \text{if } p_2 \leq p_1 - \frac{s}{q} \quad (\text{Search Activated}) \end{cases} \quad (4)$$

For Firm 2, the niche demand with mass $q(1 - q)$ is always present irrespective of search. When search is activated, Firm 2 gets an additional demand of q^2 , those with good draws for both firms and switching to Firm 2 after searching. For Firm 1, when search is deterred, all matched consumers buy from Firm 1 immediately. When search is activated, Firm 1's demand consists only of returning demand who finds $V_2 = 0$.

3 Equilibrium of the Pricing Game

The discontinuity in demand at $p_1 - p_2 = \frac{s}{q}$ and the flatness in demand when price gap goes strictly above or below the cutoff imply no Pure Strategy Nash Equilibrium exists. We solve for an asymmetric Mixed Strategy Nash Equilibrium where Firm j chooses p_j according to CDF $F_j(p)$ over support $[\underline{p}_j, \bar{p}_j]$, with no interior mass points or gaps. The asymmetry reflects the informational advantage of Firm 1 over Firm 2 imposed by the difference in prominence. We conjecture this equilibrium is unique since any mass or gap at the interior will unravel an MSNE.

Step 1: Equilibrium Profits and Upper Bounds. The upper bounds are equal to the reservation prices. See Proposition 1 in Section 3 for the proof.

- Firm 1: The upper bound is $\bar{p}_1 = V$. At this price, Firm 1 sells only to the returning demand (who return after a failed search). The equilibrium profit is anchored at the top with the “safe” demand $q(1 - q)$:

$$\bar{\Pi}_1 = Vq(1 - q).$$

- Firm 2: The upper bound is limited by the participation of consumers who found $V_1 = 0$. Thus, $\bar{p}_2 = V - \frac{s}{q}$. At this price, Firm 2 sells only to the niche demand. Again, the equilibrium profit is anchored at the top with the “safe” demand:

$$\bar{\Pi}_2 = (V - \frac{s}{q})q(1 - q).$$

Step 2: Deriving the CDFs. We use the indifference conditions: expected profit at any price p in the support must equal the equilibrium profit.

Firm 1’s mixing (F_1) is derived from Firm 2’s indifference: Firm 2 earns $\bar{\Pi}_2$ by balancing between activating search and exploiting the niche demand. Search is activated iff $p_1 \geq p_2 + \frac{s}{q}$.

$$p_2[q(1 - F_1(p_2 + \frac{s}{q})) + q(1 - q)F_1(p_2 + \frac{s}{q})] = \bar{\Pi}_2$$

Solving for $F_1(p)$ (letting $p = p_2 + \frac{s}{q}$):

$$F_1(p) = \frac{1}{q} \left[1 - \frac{(V - \frac{s}{q})(1 - q)}{p - \frac{s}{q}} \right]. \quad (5)$$

Firm 2’s mixing (F_2) is derived from Firm 1’s indifference: Firm 1 earns $\bar{\Pi}_1$ by balancing between deterring search and exploiting the returning demand. Search is deterred iff $p_2 > p_1 - \frac{s}{q}$.

$$p_1[q(1 - F_2(p_1 - \frac{s}{q})) + q(1 - q)F_2(p_1 - \frac{s}{q})] = \bar{\Pi}_1$$

Solving for $F_2(p)$ (letting $p = p_1 - \frac{s}{q}$):

$$F_2(p) = \frac{1}{q} \left[1 - \frac{V(1-q)}{p + \frac{s}{q}} \right]. \quad (6)$$

Step 3: Lower Bounds and Mass Points. We solve for the lower bounds of the supports, denoted $\underline{p}_1$ and $\underline{p}_2$. For Firm 1, setting $F_1(\underline{p}_1) = 0$:

$$1 - \frac{(V - \frac{s}{q})(1-q)}{\underline{p}_1 - \frac{s}{q}} = 0 \implies \underline{p}_1 = (1-q)V + s. \quad (7)$$

For Firm 2, solving $F_2(p) = 0$ yields a lower bound that violates profit feasibility. This “fake” lower bound is too low that Firm 2 would never want to mix on. To see this, derive this bound $\tilde{p}_2$:

$$F_2(\tilde{p}_2) = 0 \implies \tilde{p}_2 = (1-q)V - \frac{s}{q}$$

The maximal possible profit Firm 2 could achieve (i.e. even if demand is q) is:

$$\pi_2^{max}(\tilde{p}_2) = q(1-q)V - s < q(1-q)V - (1-q)s = \bar{\Pi}_2$$

Instead, Firm 2’s lower bound $\underline{p}_2$ is determined by the price where capturing the maximum possible demand (q) yields exactly the equilibrium profit $\bar{\Pi}_2$:

$$\begin{aligned} \underline{p}_2 \cdot q &= \bar{\Pi}_2 = q(1-q)(V - \frac{s}{q}) \\ \underline{p}_2 &= (1-q)(V - \frac{s}{q}). \end{aligned} \quad (8)$$

Since $F_2(\underline{p}_2) > 0$ (calculated via the CDF derived in Step 2), Firm 2 has a **mass point** at $\underline{p}_2$. This mass is:

$$F_2(\underline{p}_2) = \frac{s}{q((1-q)V + s)} \quad (9)$$

An Example Suppose $q = 0.5$, $V = 100$, $s = 10$. Then the pricing bounds become $\underline{p}_2 = 40$, $\underline{p}_1 = 60$, $\bar{p}_2 = 80$, $\bar{p}_1 = 100$. The mass at $\underline{p}_2 = 40$ is $\frac{1}{3}$. Therefore, Firm 2 sometimes commit to the lowest fare-sale price to activate search. See Figure 1 for the graph of the distribution functions.

4 Verification of Equilibrium Properties

We perform three specific verifications to ensure the consistency of the supports and the validity of the mass point derived above.

Verification 1: Demand Consistency at $\underline{p}_2$ (Guess and Verify). In Step 3, we assumed that at $\underline{p}_2$, Firm 2 captures demand q . This requires that $\underline{p}_2$ is low enough to activate search against *all* prices in Firm 1's support. Specifically, we need $\underline{p}_2 \leq \underline{p}_1 - \frac{s}{q}$. Checking the derived values:

$$\begin{aligned}\underline{p}_1 - \frac{s}{q} &= (1 - q)V + s - \frac{s}{q} \\ &= (1 - q)V - \frac{s}{q}(1 - q) \\ &= (1 - q)(V - \frac{s}{q}) = \underline{p}_2.\end{aligned}$$

The condition holds with equality: $\underline{p}_1 = \underline{p}_2 + \frac{s}{q}$. This confirms that at its mass point, Firm 2 is exactly on the margin of undercutting Firm 1's entire distribution. The structural identity of the equilibrium is thus $\underline{p}_1 - \underline{p}_2 = \frac{s}{q}$. Also, since $\bar{p}_1 - \bar{p}_2 = \frac{s}{q}$, the length of the two pricing ranges are the same.

Verification 2: Mass Condensing Does Not Violate Indifference Conditions. We derived $F_2(p)$ assuming a continuous distribution, but established that Firm 2 must condense probability mass into an atom at $\underline{p}_2$. We must verify that this condensation does not alter Firm 1's expected profit. Firm 1's profit depends on the cumulative probability $F_2(p_1 - \frac{s}{q})$. Due to the identity derived in Verification 1 ($\underline{p}_1 - \frac{s}{q} = \underline{p}_2$), for any p_1 in Firm 1's support ($p_1 \geq \underline{p}_1$), the relevant cutoff price satisfies:

$$p_1 - \frac{s}{q} \geq \underline{p}_2.$$

This implies that Firm 1's pricing decisions are always evaluated against values of F_2 at or above the mass point $\underline{p}_2$. Firm 1 never sets a price low enough to compete with the hypothetical "tail" of Firm 2's distribution below $\underline{p}_2$ (the part that was condensed). Thus, condensing that tail into a mass point at $\underline{p}_2$ maintains the correct probability mass for all p_1 in Firm 1's support, and Firm 1's indifference condition remains valid.

Verification 3: Pricing Upper Bounds Have No Slackness We mentioned in step 1 of Section 2 that firms' pricing upper bounds must be at their respective reservation prices. This is due to their incentive to exploit the captive segment of demand, once the search condition becomes locally fixed. The following proposition substantiates this claim.

Proposition 1. Pricing Upper Bounds.

In any MSNE where both firms' pricing functions are atomless in the interior and the upper bounds, and have connected support, it must be that $\bar{p}_1 = V, \bar{p}_2 = V - \frac{s}{q}$.

Proof. Obviously, firms will not set the upper bounds strictly above their reservation prices, which gives a demand of 0. Hence, $\bar{p}_1 \leq V, \bar{p}_2 \leq V - \frac{s}{q}$. We first show that the gap between the upper bounds must be exactly the cutoff $\frac{s}{q}$, and then show that they cannot both be lower than their

reservation prices even when the gap is equal to the cutoff (i.e. no double slackness).

First, suppose $\bar{p}_1 - \bar{p}_2 < \frac{s}{q}$. Then, any $p_2 \in [\bar{p}_1 - \frac{s}{q}, \bar{p}_2]$ (which is a non-trivial range) will make search deterred, leaving Firm 2 only with the niche demand (who finds $V_1 = 0$), $q(1 - q)$. Since Firm 2's demand is insensitive to price change (as long as $p_2 \leq V - \frac{s}{q}$), Firm 2 has an incentive to deviate by condensing all its weights in this range to $\bar{p}_2$, fully extracting surplus from the niche demand. This weight must exist because the distribution has no gaps or holes by assumption. This violates Firm 2's optimality.

Similarly, suppose $\bar{p}_1 - \bar{p}_2 > \frac{s}{q}$. Then, any $p_1 \in [\bar{p}_2 + \frac{s}{q}, \bar{p}_1]$ (which is a non-trivial range) will make search activated, leaving Firm 1 only with the returning demand, $q(1 - q)$. Due to the insensitivity of demand to price, Firm 1 would strictly profit by condensing all weight in this range to $\bar{p}_1$, fully extracting surplus from the returning demand. Again, this weight must be strictly positive. This violates Firm 1's optimality. Combined, it must be that $\bar{p}_1 - \bar{p}_2 = \frac{s}{q}$.

Finally, suppose $\bar{p}_1 = V - \epsilon$ and $\bar{p}_2 = V - \frac{s}{q} - \epsilon$ for some $\epsilon > 0$, so $\bar{p}_1 - \bar{p}_2 = \frac{s}{q}$ holds. At $p_2 = \bar{p}_2$, since F_1 has no mass at $\bar{p}_1$, search is deterred with probability 1, leaving Firm 2 only with the niche demand $q(1 - q)$. Firm 2's equilibrium profit is then $q(1 - q)(V - \frac{s}{q} - \epsilon)$ because any p_2 in the pricing range must yield this profit. However, playing a pure strategy of $p_2 = V - \frac{s}{q}$ would result in a higher price with no change in demand, profiting $q(1 - q)(V - \frac{s}{q})$, which is strictly higher. (If consumers' tie-breaking is not assumed to be in favor of Firm 2, pick a deviating $p_2 = V - \frac{s}{q} - \eta$ for some $0 < \eta < \epsilon$.) This violates Firm 2's optimality. (Similar argument works for constructing a profitable deviation for Firm 1 instead.)

Therefore, it must be that $\bar{p}_1 = V, \bar{p}_2 = V - \frac{s}{q}$.

□

5 The Effect of Prominence and Search Cost on Prices

A key finding of this model is the pricing hierarchy between the prominent firm (Firm 1) and the non-prominent firm (Firm 2). We establish that the prominent firm prices higher in the sense of First-Order Stochastic Dominance (FOSD).

Proposition 2. Price Hierarchy.

In the MSNE characterized by (5)-(9), Firm 1's price distribution first-order stochastically dominates Firm 2's price distribution. In fact, $F_1(p) < F_2(p) \forall p \in [\underline{p}_2, \bar{p}_1)$, and $F_1(\bar{p}_1) = F_2(\bar{p}_1) = 1$.

Proof. The non-overlapping regions at the two ends are straightforward. For the overlap $[\underline{p}_1, \bar{p}_2]$ derived above, it's sufficient to show that $F_1(p) < F_2(p) \forall p$ in this range.

$$F_1(p) < F_2(p) \iff p < 2V - \frac{s}{q}$$

after some algebra, which is always true because $p \leq V - \frac{s}{q}$ in this range and $V > 0$. □

Relation to the Literature. This result—that the prominent firm charges higher prices than the non or less-prominent firm(s)—contrasts with the finding in several seminal papers that argue that prominent firms often price *lower* to discourage consumers from searching further. These include Armstrong and Zhou (AZ 2011), and Armstrong et al (AVZ 2009), among others. We suspect that the divergence arises from the distinct assumptions on market and information structures, such as price observability and match value distribution.

AVZ 2009 finds that the single prominent firm charges a lower equilibrium price than the remaining finitely many firms that are equally non-prominent. In their setting, consumers always start examining the prominent firm, and if not satisfied, search randomly among the remaining firms. The main difference with ours is that they assume both hidden price and hidden product fitness, while ours only assume the latter and instead consider advertised price. Their rationale is that the prominent firm faces “fresh” demand which is more elastic, while later firms face “niche” demand (consumers who rejected the first option) which is less elastic. This incentivizes the prominent firm, which is essentially competing with the entire market, to price low to convert traffic immediately.

While AVZ 2009 treats firm prominence as exogenous, AZ 2011 considers how firms actively compete to become prominent, by incurring a cost to post their prices on a price comparison website. Under this advertised price setting, if a firm’s price happens to be the lowest among those posted on the website, this firm attracts all the price-seeker consumers who always buy the cheapest product, while other losing firms (including the price-posting and non-posting ones) only serve their respective loyal consumers. In their demonstrating model with two firms, match values are perfectly negatively correlated across firms, an assumption that is different from our i.i.d setting. This learning aspect changes consumers’ search problem. Their Hotelling-style horizontal differentiation is also less sharp than our binary match value setting.

The main result of AZ 2011 is that a higher search cost leads to lower prices. On the other hand, our comparative statics of search cost on prices is mixed. From expressions of our lower and upper bounds (Equation(7), (9), and Proposition 1), it’s easy to see that when s increases (still satisfying $s < qV$), the prominent firm’s pricing range moves upward, while that of the non-prominent firm moves downward. As search friction increases, Firm 1 enjoys more market power endowed by the position while Firm 2 has to price even lower to attract visits.

Our initial stage of position auction via Generalized Second Price (GSP) auction is indebted to Anderson and Renault (2025). Our experimental inspiration is from Davis and Holt (1996). The model closest to ours is Xu et al (2008).

Future Plan

- Theory: Uniqueness of my MSNE (why it has to have connected support and atomless interior). Intuition of the price hierarchy and comparative statics in search cost. Comparison

with broad literature.

- Experimental Design: Low vs. High search cost. Potentially also include hidden vs. advertised prices; random exogenous position allocation vs. GSP auction.
- Policy Implication: Welfare effect of introducing prominence auction (the exogenous vs. endogenous position allocation angle in the experiment). Since firms are ex-ante the same and pricing strategies only depend on realized position but not on how firms get these positions, it could be that the auction is just to shift profits from sellers to auctioneer/platform without affecting consumer surplus. Would that still be interesting?
- Behavioral Forces: One behavioral force in the lab that might confound the interpretation of pricing is sunk cost fallacy. Firm 1 participants might charge higher p_1 than their Firm 2 counterparts not by equilibration, but to compensate for their higher sunk bid in the auction stage. IO economists do not believe firms make mistakes. The push-back is that consumers' behavioral noise (e.g. quantal response in the search problem) might soften firm's undercutting incentives and fundamentally change the equilibrium pricing and bidding.

6 References

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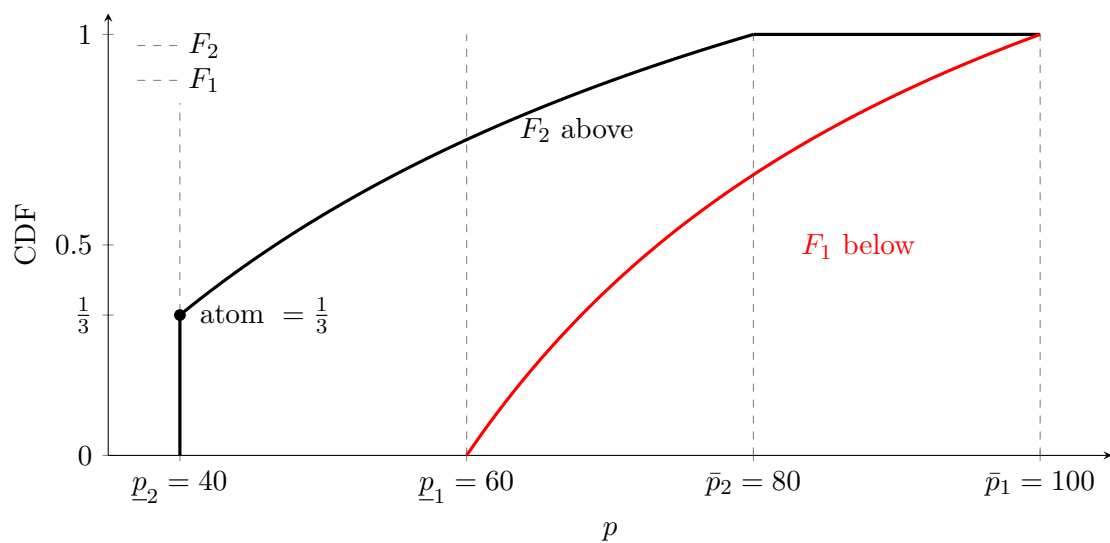


Figure 1: Equilibrium CDFs (example: $q = 0.5$, $V = 100$, $s = 10$). Here $F_1(p) \leq F_2(p)$ for all p , with strict inequality for $p < 100$, implying Firm 1 prices higher in the sense of FOSD.